SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

<u>Analytical Problem Solving</u>

Code of Conduct:

I Thanish (192412024) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Thanish

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Solve the 6 - Queens Problem : place 6 queens on a 6x6 chessboard such that no two queens attack each other. Formulate the Problem components (state space, initial state, successor function, goal test) and explain how incremental formulation reduces the search space compared to complete - state formulation

<u>Solution</u>

Problem Formulation:

a) state Space:

The space state consists of all Possible configurations of placing 0 to 6 queens on a 6x6 chessboard, ensuring that each queen occupies a different row.

b) Initial State:

* Empty 6x6 Chessboard

* No queens are placed.

c) Successor Function:

* place one queen in the next row.

* choose only columns where the queen is not attacked by Previously placed queens.

* continue until all six queens are placed.

d) Goal Test:

The goal is reached when:

* Exactly 6 queens are placed.

* No two queens attack each other (same row, same column, or diagonal).

One valid Solution

Row	column
1	2
2	4
3	6
4	1
5	3
6	5

Block Representation:

```
. Q . . . .
. . . Q . .
. . . . . Q
Q . . . . .
. . Q . . .
. . . . Q .
```

(Queen = Q)

Incremental formulation vs complete state formulation

Complete state formulation

- [illegible] all 8 queens at once
- [illegible] some [illegible] arrangement
- number of [illegible] is very large
- Many invalid states are generated

Incremental formulation

- Places one queen at a time
- Immediately [illegible] [illegible] [illegible] [illegible]
- Only valid partial states are [illegible]
- Greatly reduces the search space

[illegible]

Complete state	Incremental
Places all queens together	Places one queen at a time
Large search space	Small search space
Many invalid states	Invalid states pruned early
Less efficient	More efficient

2] For a chess-playing agent, characterize the task environment across all PEAS/ environment dimensions (observable, deterministic, episodic, static, discrete, agents Justify each classification with reference to the rule of chess.

Solution:

PEAS

Performance Measure:

* Win the game
* Checkmate the opponent
* Minimize mistakes.
* Maximize strategic advantage

Environment:

* Chessboard
* Pieces
* opponent

Actuators:

* Move chess pieces

Sensors:

* observe board positions
* Detect legal moves.
* Detect opponent moves.

Environment characteristics

1. Observable

Fully observable

Reason:

* Both players can see the complete chessboard

* No hidden information exists

2. Deterministic

Deterministic

Reasons:

* Every move has a predictable outcome

* No randomness or chance is involved

3. Episodic or Sequential

Sequential

Reason:

* Every move affects future moves.

* current decisions influence the entire game.

4. static or Dynamic

static

Reason:

* The board does not change while player is thinking

* Time does not alter the board state.

5. Discrete or continuous

Discrete

Reason:

* Finite number of legal moves.
* Board has fixed squares.

6. Single-Agent or Multi-Agent

Multi Agent

Reason:

* Two intelligent players compete
* one player's actions affect the other.

Summary Table

Property	Classification	Reason
observable	Fully observable	Entire board visible
Deterministic	Deterministic	No randomness
Episodic	Sequential	Moves affects future
static	static	Board unchanged while thinking
Discrete	Discrete	Two competing players
Agents	Multi-Agent	Finite legal moves

An exploration robot is sent into an unknown cave system with no prior map, & its sensors have limited range. Explain how the robot can use search with partial information (contingency / online search) to build up a map while searching for exist, & discuss the risks of irreversible actions in this environment.

solution :

Problem :

The robot.

* Has no map.

* sensors have limited range

* Must search for the exit while building a map.

online (contingency) search

Since the robot has incomplete information, it performs online search.

step 1 :
start from the entrance

step 2 : sense nearby passages using sensors

step 3 : update an internal map with discovered locations

step 4 : choose an unexplored path

step 5 : Repeat until the exist is found.

As exploration continues, the map becomes more complete

Building the Map:

The robot records:

* Visited locations
* unexplored paths
* obstacles
* Dead ends
* This gradually creates a complete map of the cave

Contingency Search:

Because information is uncertain;

* The robot continuously sense the environment.
* It changes its plan based on newly discovered paths
* It can backtrack if necessary.

Risks of Irreversible Actions

Some actions cannot be undone.

Examples :

* Falling into a deep pit
* Entering a collapsing tunnel.
* Crossing a one-way Passage
* Running out of battery after taking a wrong route

These may Prevent the robot from reaching the exit.

How to Reduce Risks

* Explore cautiously
* Avoid unknown dangerous path
* keep track of visited locations
* Reverse battery power.
* Backtrack whenever possible before entering risky areas

Conclusion:

An exploration robot uses online (contingency) search to explore an unknown cave, gradually builds a map from sensor data, and continuously its decisions based on new information, carefull exploration & avoiding irreversible actions help improve the chances of safely finding the exist.